AI Career Path Finder

Abstract

The **AI Career Path Finder** is a **Python-based career guidance system** designed to help freshers explore suitable career opportunities using information from their **resumes, skills, education, experience, and interests**.

The system extracts skills from uploaded resumes and combines them with manually selected skills. It compares the user's skill profile with career-role requirements and uses selected interests to identify relevant career options. The system presents **career recommendations**, highlights **existing skills**, and identifies **skills that need to be developed** for different roles.

Users can select a **target career** to receive a detailed analysis of their current skill profile. The system identifies **skill gaps** and generates a **personalized career roadmap** containing recommended skills and technologies, project ideas, practice tasks, and interview preparation topics.

Users can update their skills and profile information and generate an updated analysis and roadmap. The application also supports **saved profiles, analysis history, roadmap progress tracking, and career comparison**.

The project aims to help freshers **understand their career options, identify areas for improvement, and follow a structured learning path** toward their desired career roles.